Xingdong Zuo

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Interests

Offline Reinforcement Learning for Recommender Systems and Auto-Bidding, Robot Learning with Vision-Language-Action Models, World Action Models

Experience

May 2021 – present **Research Engineer**, Naver Corp – South Korea

Research and development of real-world reinforcement learning algorithms

- Developed auto-bidding for display ads using BC and IQL (offline RL), achieving significant improvement in ads efficiency; presentation at Naver DAN 2025.
- Improved CVR and ROAS of NAVER Dynamic AD with advanced retrieval and RL models.
- Developed RL recommendation models for LINE Wallet, boosting long-term user engagement.
- Integrated offline RL into ranking models for Smart Message.
- Enhanced embedding and reranker models for Brand Chatbot and GAIA Shopping.
- Prototyped RLHF for user foundation models.
- Co-authored a NeurIPS 2023 publication and the HyperCLOVA X Technical Report; filed a Korean patent (2025).
- Built QueryLake ETL library and enhanced KPI reporting pipelines.

May 2017 – Nov 2017 **Master Thesis**, Bosch Center for Artificial Intelligence – Germany

Title: Sample-efficient model-based policy search via Bayesian Neural Networks Dynamics Models

- Achieved better sample efficiency: from ~10 to ~6 trials

Education

2015 – 2019 **University of Freiburg**, M.Sc. in Computer Science – Freiburg, Germany
Focus: Machine Learning and Reinforcement Learning

2010 – 2015 **Linnaeus University**, B.Sc. in Applied Mathematics – Sweden
Minor: Computer Science

Skills

AI Stack: Claude Code, PyTorch, Polars, Kubeflow, Docker, Trino SQL, Qdrant, Linux, Git

Robotics: LeRobot, Vision-Language-Action models (SmolVLA), ACT, SO-101 / LeKiwi platforms

Languages: English (fluent), Chinese (native)

Publications

2026 **Constrained Auto-Bidding via Generative Response Modeling**
E Yang, X Zuo, K M Kim
arxiv.org/abs/2605.27811 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD))

2026 **Drift-VLA: Fast Vision-Language-Action Policies with One-Step Drifting**
X Zuo
zuoxingdong.github.io/drift-vla (Preprint)

2025 **Method and System to Provide Information Data with Minimal Infrastructure Resources**
X Zuo, et al.
patents.google.com/patent/KR20250124713A/en (Korean Patent Publication No. 10-2025-0124713)

2024 **HyperCLOVA X Technical Report**
HyperCLOVA X Team (incl. X Zuo)
arxiv.org/abs/2404.01954 (arXiv preprint)

2023 **Direct preference-based policy optimization without reward modeling**
G An, J Lee, X Zuo, N Kosaka, K M Kim, H O Song
arxiv.org/abs/2301.12842 (Advances in Neural Information Processing Systems 36)

2021 **Is Learning World Model Always Beneficial For Reinforcement Learning?**
X Zuo
10.13140/RG.2.2.33768.80641 (Technical report)

2014 **Numerical simulation of Asano-Khrennikov-Ohya quantum-like decision making model**
X Zuo
(NeuroQuantology 12 (4))

Talks

2025 **Bid, Win, Profit: Revenue Optimization with a Generative-AI Ad-Auction Simulator** – **Seoul**
Invited talk at DAN 25, NAVER's annual tech conference (recording).

Projects

2026 – present **Mobile Manipulation with LeKiwi and PincOpen — Hugging Face**
End-to-end mobile manipulation on LeKiwi (SO-101 arm, holonomic base); workflow published as a Hugging Face blog.

- Fine-tuned SmolVLA for pick-and-place with data-scaling and vision-encoder ablations; the unfrozen encoder learned base navigation from only 10 of 130 episodes.
- Integrated PincOpen gripper, upgraded servos, and fixed an arm-extension voltage-drop failure via P-gain tuning.
- Published reproducible SmolVLA evaluation on LIBERO-Spatial (blog).

2025 – 2025 **LeRobot Worldwide Hackathon 2025 — Hugging Face**

Led a 5-person team over 2 days to complete a language-instructed manipulation task using SmolVLA.

- Fine-tuned SmolVLA for language-instructed color-ball pickup on real hardware
- Led team coordination, task decomposition, and end-to-end workflow

2025 – 2025 **LeRobot (contributor) — Hugging Face**

Contributions to video encoding and CLI tools.

- Authored a merged feature for batched video encoding in dataset recording, adding a video_encoding_batch_size option plus cleanup and exception handling for interrupted sessions (PR #1390).

2025 – present **lerobokinson & lekiwi-tui (author)**

GUI for professional PD gain tuning of Feetech/Dynamixel servos; terminal control for LeKiwi robots.

2025 – present **rlbidder (author)**

Modular reinforcement learning auto-bidding library for research and production.

2018 – 2020 **OpenAI Gym (top #5 contributor, Nov. 2019)**

42 merged PRs on wrappers and vectorized-environment infrastructure.

- Implemented 12 (out of 14) different Wrappers classes
- Refactored spaces and add new features: flatdim, flatten, unflatten

2018 – 2020 **PyTorch (contributor)**

- Implemented: TanhTransform, DataLoader(drop_last), convert Module params, ModuleList.insert
- Improved numerical stability in SigmoidTransform

2018 – 2021 **lagom (author)**

A PyTorch Library for Rapid Prototyping of Reinforcement Learning Algorithms (378★)

2018 – 2020 **mazelab (author)**

A customizable gym environment for maze/gridworld (271★, 13 citations).

2017 – 2017 **Value Iteration Networks in PyTorch and TensorFlow**

Widely used implementations for follow-up research (223★).

Other open-source libraries (author)

dm2gym (87★, DeepMind Control Suite wrapper), DeepPILCO (54★, model-based RL with uncertainty), recsys_metrics (28★), gym-recsys (26★)